

Predicting local laser-track geometry from in-situ melt-pool thermal sequences

Final report, NSF Future Manufacturing Data Challenge (dataset arXiv:2607.07965)

Development tracks 8/10/14; held-out test track 21

1 Problem setup

We work on a 0.2 mm grid along the scan coordinate x over the common 20–100 mm window (400 positions per track, matching the 0.2 mm/frame spacing of the 50 fps thermal stream). Given thermal frames T and the position x , we predict a vector $g(x)$ of local geometry descriptors as five conditional quantiles,

$$\hat{F}_{g(x)}^{-1}(q \mid T_{x-k:x+k}, x), \quad q \in \{0.05, 0.25, 0.50, 0.75, 0.95\}, \quad (1)$$

where g contains the width $w(x)$, the boundaries $y_L(x)$, $y_R(x)$, the center $c(x)$, edge roughness, waviness, contour deviation, peak relief, cross-sectional area, and the profile $z(y \mid x)$. Height maps are used for targets only; every model input is either measured during the scan or is the coordinate x . All settings below were fixed by leave-one-track-out (LOTO) cross-validation on the development tracks before track 21 was evaluated once.

2 Targets from the height maps

The remelted track is optically smooth while the machined substrate is rough, so the interferometer returns dense valid data inside the track and sparse data outside. The per- y valid-pixel fraction $v(y)$ within a bin is therefore close to binary. With the threshold

$$\tau = \frac{1}{2} (P_{10}[v] + P_{90}[v]), \quad (2)$$

the boundaries $[y_L, y_R]$ are the sub-pixel edges of the largest run $\{y : v(y) > \tau\}$, giving $w = y_R - y_L$ and $c = (y_L + y_R)/2$, with per-bin checks on contrast, width range, edge contact, and coverage. This keeps 362 of 400 track-21 bins despite the partial profilometry failure on that track. Writing S_L for a moving-average smoother of window L and $b \in \{y_L, y_R\}$ for a boundary series,

$$r(x) = \text{std}_{\pm 1\text{mm}}[b - S_2 b], \quad \omega(x) = \text{std}_{\pm 5\text{mm}}[S_2 b - S_{10} b], \quad d(x) = |b - (\alpha + \beta x)|, \quad (3)$$

averaged over both edges (roughness, waviness, and deviation from a per-track linear toolpath fit). Cross-sections are detrended per bin with a linear fit to the two substrate margins before we extract the peak relief h_{p90} and the area $A = \int z \, dy$.

3 Thermal features and the isotherm anchor

For each frame f (400×400 px, $14 \mu\text{m}/\text{px}$) we compute melt-pool descriptors (area, robust extents, elongation, centroid, cooling-tail energy, scan asymmetry) with ± 0.4 and ± 2 mm rolling statistics, and the cross-track extent of the c -count isotherm,

$$E_c(x) = \Delta_{\text{px}} (\max \mathcal{J}_c - \min \mathcal{J}_c), \quad \mathcal{J}_c = \{j : \exists i, f_{ij}(x) > c\}, \quad (4)$$

for $c \in \{1300, \dots, 1800\}$. The deposited width is set by the solidification isotherm, so at the right count E_c should track width independently of laser power. On the development tracks the ratio w/E_{1400} is 0.887, 0.839, and 0.831 (CV 2.9%; picking c by minimum ratio CV over the development tracks selects $c^* = 1400$). The anchor is

$$a(x) = k^* \bar{E}_{1400}(x), \quad k^* = \frac{1}{3} \sum_{t \in \text{dev}} \frac{\text{med}[w_t]}{\text{med}[\bar{E}_{1400,t}]} = 0.853, \quad (5)$$

where $\bar{E}$ is the ± 2 mm rolling mean. Track 21 sits far outside the development range (hot area 0.027 vs. 0.28–0.46 mm²); trees saturate there, a linear anchor does not.

4 Residual quantile model, calibration, and testing

The width model keeps the extrapolating level separate from the learned part:

$$\hat{w}_q(x) = a(x) + G_q(\phi(x)), \quad G_q = \arg \min_G \sum_{x \in \text{dev}} \rho_q(w(x) - a(x) - G(\phi(x))), \quad (6)$$

where $\phi(x)$ stacks the thermal descriptors and G_q is a gradient-boosted tree ensemble (150 trees, depth 2, LOTO-selected) trained with the pinball loss $\rho_q(u) = u(q - \mathbf{1}[u < 0])$; predicted quantiles are sorted. Feeding $a(x)$ to a standard GBM as one more feature does not work (MAE 0.260 mm), because trees cannot extrapolate below their training range; the split in (6) is what makes the anchor useful. The level uncertainty for an unseen power setting is estimated from the three LOTO fold errors $e_j = \text{med}[w] - \text{med}[\hat{w}_{0.5}]$, $s = \text{RMS}(e_j) = 0.052$ mm, and folded into the quantiles:

$$\tilde{Q}(q) = Q(0.5) + \text{sign}(z_q) \sqrt{(Q(q) - Q(0.5))^2 + (z_q s)^2}, \quad z_q = \Phi^{-1}(q). \quad (7)$$

We score with MAE, 90% interval coverage, and a CRPS approximation,

$$\text{CRPS} \approx \frac{2}{q_{\max} - q_{\min}} \int_{q_{\min}}^{q_{\max}} \rho_q(y - Q(q)) dq \quad (\text{trapezoid over the five quantiles}). \quad (8)$$

Models are compared with a paired moving-block bootstrap of per-bin absolute errors (block = 25 bins = 5 mm, 2000 resamples). The criterion, fixed in advance: a contribution counts as confirmed only if the 95% CI lower bound of ΔMAE is above zero and the relative improvement is at least 5%. The anchor idea carries over to the height direction, $h_{p90} \approx k_h \bar{E}_{1400} + G(\phi)$ with $k_h = 7.27 \mu\text{m}/\text{mm}$ and $A \approx k_a \bar{E}_{1400}^2 + G(\phi)$ with $k_a = 3.39$. Full cross-sections use a functional PCA fitted on the development tracks,

$$z(u | x) \approx \mu(u) + \sum_{k=1}^3 c_k(x) \varphi_k(u), \quad u = \frac{y - y_L}{w} \in [0, 1], \quad (9)$$

with the coefficients $c_k(x)$ regressed from $\phi(x)$.

5 Results (one frozen track-21 evaluation, 362 bins)

Descriptor	MAE (best)	vs. x-only	90% cov.	Verdict
Width $w(x)$	0.089 mm	−75.6%, CI [0.246, 0.305] mm	0.88	confirmed
Boundaries y_L / y_R	0.092 / 0.067 mm	−58% / −62%	—	confirmed
Waviness $\omega(x)$	0.0035 mm	−39%	0.93	confirmed
Peak relief h_{p90}	2.15 μm	−55% (tree 3.17)	0.77	confirmed
Area $A(x)$	0.96 $\mu\text{m} \cdot \text{mm}$	−67% (tree 2.04)	0.90	confirmed
Center $c(x)$	0.062 mm	x-only already best	0.98	no thermal gain
Edge roughness $r(x)$	0.0132 mm	no gain, either modality	1.00	not predictable
Contour deviation $d(x)$	0.0184 mm	$\approx$ global median	0.91	not predictable
Profile $z(y x)$	2.42 μm	shape modes fail dev-LOTO	—	amplitude only

6 Sources of variation and what the results mean

The table splits local geometry by physical pathway. Quantities driven by the melt-pool thermal history (w , y_L/y_R , ω , h_{p90} , A) are predicted well from in-situ data; the waviness gain fits a melt-pool oscillation origin. The center line needs only x , which is what one would expect for a toolpath-defined quantity. Edge roughness, contour deviation, and the ripple-scale profile shape were not predictable from either modality. We tested the substrate route directly with a substrate-only SEM protocol (track band detected and masked with a ± 0.30 mm margin, texture features taken from 1.0 mm strips on both sides, band location never used

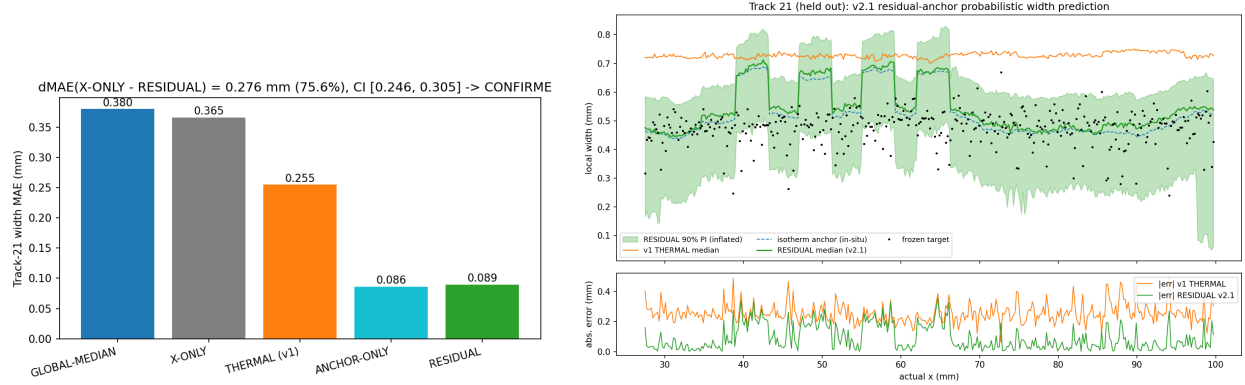


Figure 1: Left: track-21 width MAE by model; the anchor (5)–(6) removes the 0.25 mm level bias that pure tree models carry across the power gap. Right: predictions along the track with the 90% interval from (7).

as a feature): no target passed the criterion. Width gained +1.6% with CI $[0.0008, 0.0024]$ mm, above zero but below the 5% bar, close to the 1.8% unconfirmed SEM gain reported by an independent baseline on the same data.

Three points summarize the interpretation. First, one physical quantity does most of the work: the ratio $w/E_{1400} \approx 0.85$ holds across a $2\times$ power range, which points to the solidification isotherm as the mechanism that sets the width. Second, model structure matters as much as the feature itself: the same anchor fails as a tree input and works in the residual form (6). Third, the remaining along-track fluctuation appears information-limited rather than model-limited: within-track thermal–width correlations stay below $|r| = 0.35$, predicting the shape modes in (9) is worse than a constant mean shape even in development LOTO (2.04 vs. 1.89 μm), and the rank-3 reconstruction floor of 0.41 μm bounds what perfect coefficients would give. A CNN on the same inputs would face the same limit with 1,122 training rows over three power settings.

7 Limitations

Three power settings make every cross-condition calibration thin. The height anchor over-predicts in the near-threshold 200 W regime, where the relief scaling is no longer linear. The isotherm idea was checked coarsely on all four tracks before the freeze; all frozen constants come from the development tracks, but a pre-registered replication on new tracks would be the cleaner test. Scan orientation per track is inferred from development data rather than documented. Finally, results depend on the label definition: replicating the method inside the baseline repository showed that its gradient-edge labels are nearly power-invariant (0.741 to 0.804 mm from 200 to 400 W), and under that definition the anchor is correctly rejected by the same criterion, while it is confirmed (70.1% improvement) under the validity-run labels used here. The full pipeline, frozen configuration files, hashes, and a single reproduction command accompany this report.